

**Software Design & Analysis
(CS3004)**

Date: Nov 4, 2025
Course Instructor(s)
AR, MK, ZN, LK

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Roll No

Section

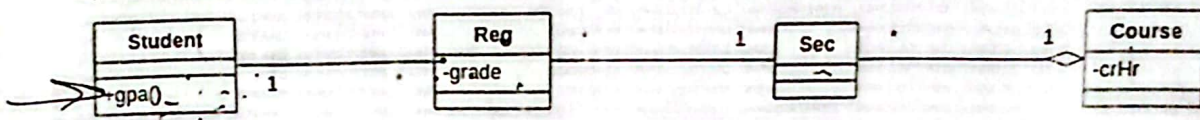
Signature

***** No separate answer sheet required! *****

CLO 3: Use different UML notations for software design

[20 marks]

Q1: Consider the following class diagram for a course registration system:



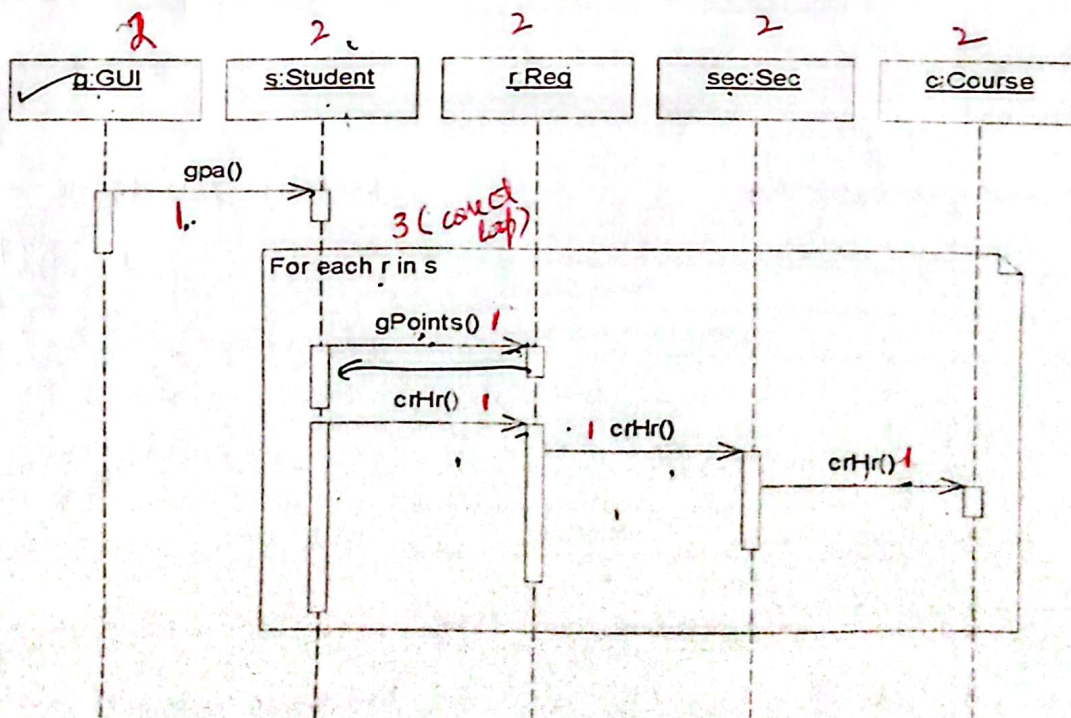
Here Reg stands for registration, Sec for section, and crHr means credit hours.

Now give a sequence diagram for the function gpa() show the object interactions required to execute this operation. You may add a few functions in the classes above.

The formula to compute GPA is as follows:

$$GPA = (gPoint1 * crHr1 + gPoint2 * crHr2 + \dots) / \text{total credit hours}$$

where gPoint refers to grade points (4 for A, 3 for B, 2 for C, 1 for D, and 0 for F)



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CLO 5: Describe software design guidelines and principles

[10+10 marks]

Q2: Task:

- 1) Identify which SOLID principle(s) is/are violated by the design.
- 2) Refactor (improve) the design in the light of SOLID principles.

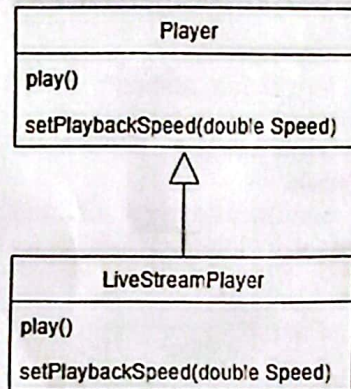
Scenario 1)

A software company is developing a Multimedia System with a class Player that represents any type of media player.

The Player class provides two basic operations:

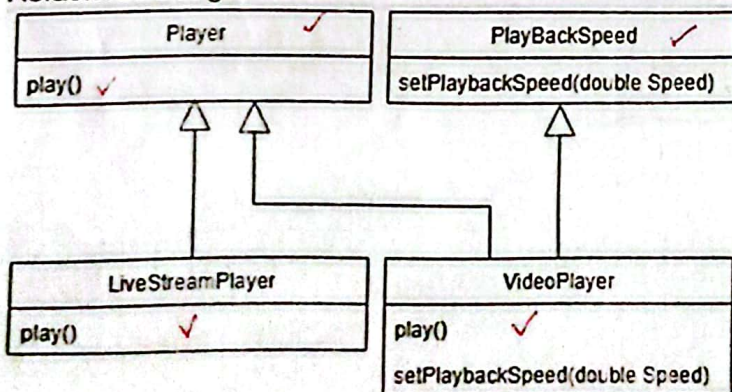
- play() to start playback
- setPlaybackSpeed(double speed) to adjust playback speed

The team currently has a LiveStreamPlayer class that plays live content. Since live content cannot change playback speed, they decided to throw an exception inside setPlaybackSpeed() for that class. In the future, they plan to add a VideoPlayer class that can both play and change playback speed.



SOLID principle violated: LSP ~~#~~

Refactored Design:



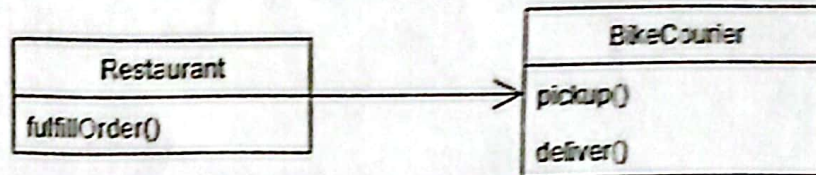
Reasons: You have a base class player with two operations:
→ play();
→ setPlaybackSpeed();

You then create a subclass "LiveStreamPlayer", but this subclass throws an exception inside "setPlaybackSpeed()" because liveStream speed cannot be changed.

Scenario 2)

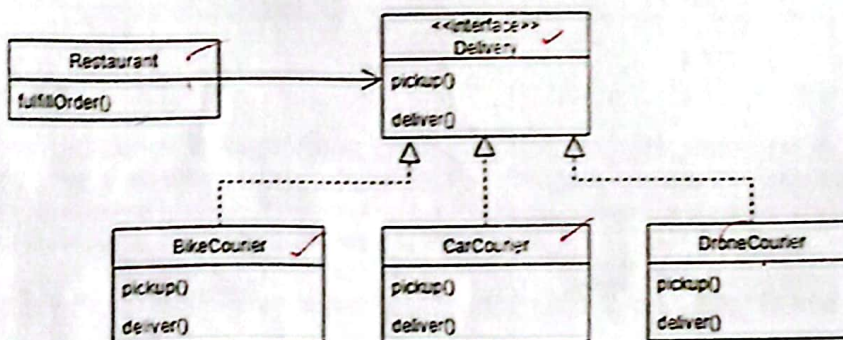
A restaurant management system needs to deliver food orders to customers. Currently, the Restaurant class uses a BikeCourier service to pick the order and deliver.

→ However, the development team plans to extend the delivery system in the future by adding new courier types like CarCourier and DroneCourier.



SOLID principle violated: DIP + OCP

Refactored Design:



Reason:

OCP →

Adding new
courier type
= extended the
system

But Restaurant
class must be modified
to support each new courier
so Restaurant is
not closed for changes.

DIP →

Restaurant directly
depends on a concrete class
not an abstraction